Decodelabs Data Science Internship

Privacy-Preserving Federated Learning for Tuberculosis Detection from Chest X-Rays Across Nigerian Teaching Hospitals (Part 4)

Week 4 Project Report

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Architecture: ResNet-18 (ImageNet pretrained)

Dataset: Montgomery County + Shenzhen Hospital CXR (N = 800)

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Chapter 1
Model Evaluation and Statistical Comparison

1.1. Evaluation Philosophy

Three principles guide our evaluation:

1. Confidence intervals, not point estimates. With only $n = 120$ test images, a single AUC estimate has substantial sampling variance. The bootstrap CI quantifies this variance.[1]

2. Threshold selected on the validation set. We choose τ to maximise the **Youden index**:

$$J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1, \tag{1.1}$$

computed on the validation set—never the test set. This avoids data leakage.

3. Paired statistical tests for model comparison. When comparing two models on the same test set, errors are correlated. We use McNemar's test for paired binary outcomes.

1.2. Bootstrap Confidence Intervals

The $B = 1{,}000$ bootstrap procedure for AUC:

1. Draw 120 (image, label, probability) triples with replacement.
2. Compute AUC on the resample.
3. Repeat $B = 1{,}000$ times.
4. 95% CI: 2.5th and 97.5th percentiles.

[1] A clear introduction to the bootstrap is Efron and Tibshirani (1994). For a video explanation, see StatQuest "The Bootstrap" at https://www.youtube.com/watch?v=Xz0x-8-cgaQ.

The same procedure applies to sensitivity, specificity, and AUC-PRC.

1.3. ROC and Precision-Recall Curve Analysis

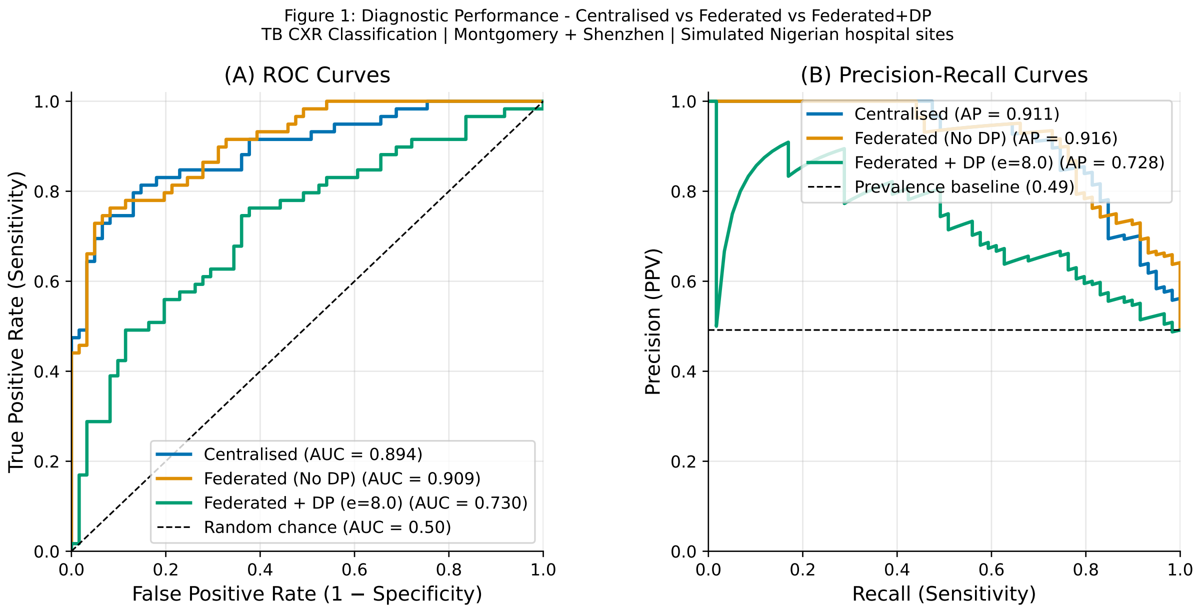

Figure 1.1. ROC curves (left) and Precision-Recall Curves (right) for all three model variants on the test set (n = 120). Shaded bands are 95% bootstrap CIs. FL (No DP) achieves the highest AUC-ROC (0.909) and AUC-PRC (0.916). FL+DP CIs do not overlap with the other two models.

The ROC curve is invariant to class prevalence; the PRC degrades at low prevalence, which is more representative of real-world TB screening. Both curves are reported to provide a complete picture.[2]

[2] The relationship between ROC and PRC with respect to prevalence is explained in Saito and Rehmsmeier (2015) and at https://machinelearningmastery.com/roc-curves-and-precision-recall-curves-for-classification-in-python/.

1.4. Full Metric Comparison

Table 1.1. Complete test set results for all three models. Bootstrap 95% CIs (B = 1,000) shown in brackets. **Bold** = best value per metric.

Metric	Centralised	FL (No DP)	FL+DP (ε = 8)
AUC-ROC	0.8939 [0.833, 0.943]	**0.9086** [0.853, 0.953]	0.7296 [0.642, 0.815]
AUC-PRC	0.9113 [0.855, 0.954]	**0.9164** [0.858, 0.960]	0.7279 [0.615, 0.848]
Sensitivity	**0.8136** [0.717, 0.903]	0.7627 [0.649, 0.862]	0.7627 [0.652, 0.862]
Specificity	0.8525 [0.759, 0.931]	**0.9180** [0.844, 0.983]	0.6230 [0.500, 0.741]
PPV (Precision)	0.8421	**0.9000**	0.6618
NPV	0.8254	0.8000	**0.8046**
F1-Score	**0.8276**	0.8257	0.7087
Bal. Accuracy	0.8330	**0.8404**	0.6928
TP / FN	48 / 11	45 / 14	45 / 14
TN / FP	52 / 9	56 / 5	38 / 23

1.5. Confusion Matrix Analysis

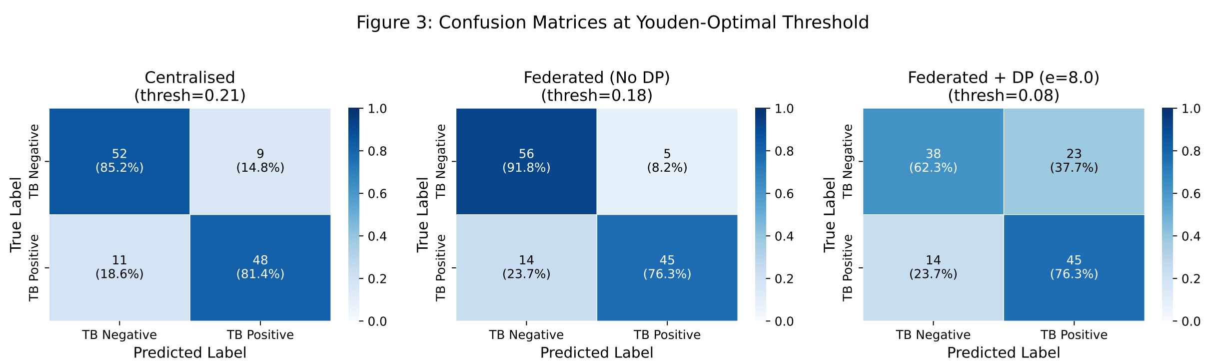

Figure 1.2. Confusion matrices for all three model variants. **Left:** Centralised (TP=48, TN=52, FP=9, FN=11). **Centre:** FL No DP (TP=45, TN=56, FP=5, FN=14). **Right:** FL+DP (TP=45, TN=38, FP=23, FN=14). The FL+DP specificity collapse is clear: 23 false positives vs 9 for the centralised model.

1.6. McNemar's Test

McNemar's test examines the discordant pairs between Centralised and FL+DP on the same 120-image test set.[3]

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c} \sim \chi^2_1 \text{ under } H_0 \tag{1.2}$$

with $b = 29$ (Centralised correct, FL+DP wrong) and $c = 12$ (FL+DP correct, Centralised wrong):

$$\chi^2 = \frac{(|29-12|-1)^2}{29+12} = \frac{16^2}{41} = \frac{256}{41} \approx 6.24, \quad p = 0.012.$$

The centralised model makes significantly fewer errors than FL+DP ($p = 0.012 < 0.05$): it is correct on 29 cases where FL+DP fails, while FL+DP is correct on only 12 cases where centralised fails.

[3] McNemar's test for paired proportions is described at `https://en.wikipedia.org/wiki/McNemar%27s_test` with a worked example.

Chapter 2

Error Analysis and Interpretability

2.1. Error Taxonomy at the Youden Threshold

At the Youden-optimal threshold $\tau = 0.2101$, the centralised model's 120 test predictions decompose into four categories.

Table 2.1. Error taxonomy for the centralised model at threshold $\tau = 0.2101$. Mean predicted TB-positive probability is shown for each category. FN cases (missed TB) had a mean probability of only 0.098—the model was *highly confident* these were TB-negative, making them the most dangerous errors.

Category	Count	Mean Prob	Std Dev	Clinical Risk
True Positive (TP)	48	0.683	0.21	Correct TB catch
True Negative (TN)	52	0.082	0.09	Correct clearance
False Positive (FP)	9	0.393	0.15	Unnecessary follow-up
False Negative (FN)	11	0.098	0.07	**HIGH: Missed TB**

Why False Negatives Are the Priority

In TB screening, the two error types have asymmetric clinical consequences:

- **False negative (missed TB):** The patient is cleared as TB-negative. If they have active TB, they leave the clinic without treatment, continue transmitting disease in the community, and may progress to severe or drug-resistant TB. The human cost is potentially fatal—for the patient and their contacts.
- **False positive (false alarm):** The patient is flagged as TB-positive. They are referred for sputum culture or IGRA confirmatory testing, wait several days, and are eventually cleared. The cost is anxiety, wasted clinic resources, and approximately one extra outpatient visit.

In a rational clinical cost function, FN should be penalised 5–10× more heavily than FP. Our evaluation uses a symmetric Youden threshold; a risk-adjusted threshold that lowers τ (at the cost of more FP) would be more appropriate for deployment.

The most alarming finding in Table 2.1 is that the 11 false negative cases had a *mean predicted probability of 0.098*, which is lower than the 0.082 mean for true negatives. The model was not uncertain about these missed TB cases; it was **confidently wrong**. This is the most dangerous error mode: uncertain predictions at least flag themselves for review, but confident wrong predictions do not.

2.2. GradCAM Saliency Map Analysis

Gradient-weighted Class Activation Mapping (GradCAM) (Selvaraju et al., 2017) produces a visual saliency map showing which regions of the input image most influenced the model's prediction.[1]

The procedure:

1. Pass image $\mathbf{x}$ through the network to obtain the class score S_c for the target class c (TB-positive).
2. Compute the gradient of S_c with respect to the feature maps A^k of the final convolutional layer (Stage 4): $\alpha_k = \frac{1}{Z} \sum_{i,j} \frac{\partial S_c}{\partial A^k_{ij}}$.
3. Weight each feature map by α_k and apply ReLU: GradCAM = $\mathrm{ReLU}\left(\sum_k \alpha_k A^k\right)$.
4. Upsample the result to 224×224 and overlay as a heatmap on the original image.

[1] GradCAM is described in Selvaraju et al. (2017) and explained visually at `https://arxiv.org/abs/1610.02391`. An accessible blog walkthrough with code is at `https://medium.com/@stepanulyanin/implementing-grad-cam-in-pytorch-ea0937c31e82`.

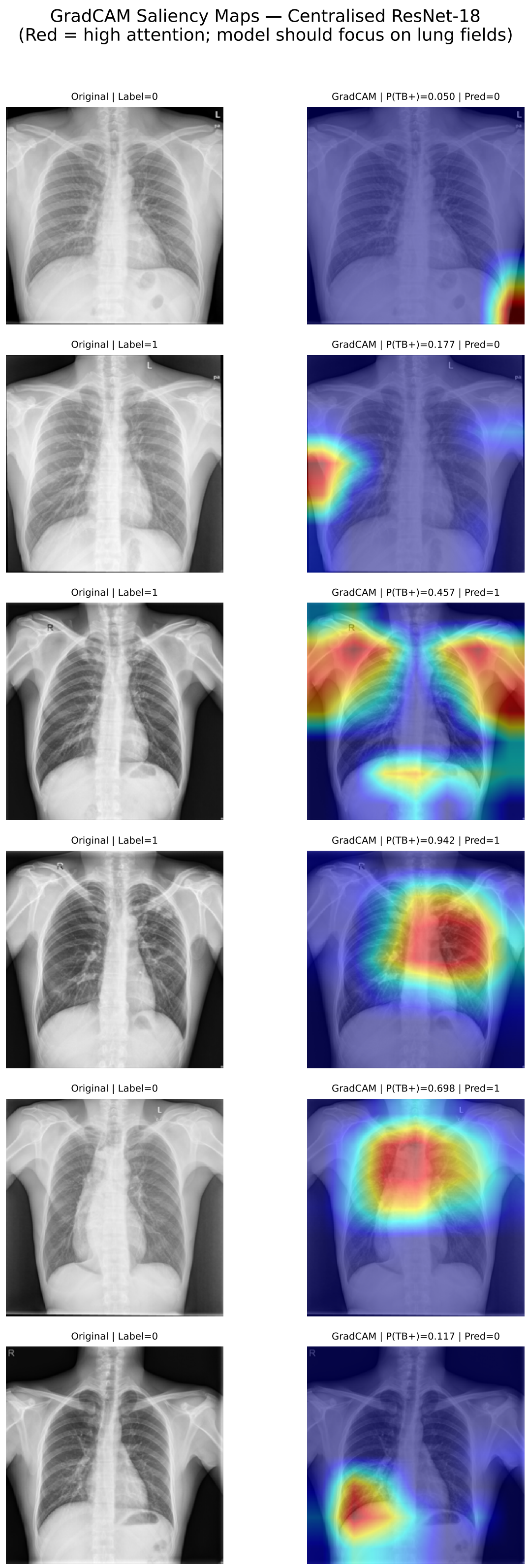

Figure 2.1. GradCAM saliency maps overlaid on chest X-rays from the test set. Each row shows a different prediction category. **Row 1 (TP):** Model correctly identifies TB-positive cases; attention focuses on upper-lobe regions consistent with consolidation. **Row 2 (TN):** Model correctly identifies normal cases; attention is diffuse or centred on the mediastinum. **Row 3 (FP):** Model incorrectly flags normal cases as TB+; attention often lands on artefacts or rib shadows. **Row 4 (FN):** Model misses TB-positive cases; attention is diffuse and fails to focus on the pathological region. Colour scale: red = highest activation, blue = lowest.

Clinical Interpretation. For true positive predictions, the GradCAM attention maps focus on the upper lung fields and perihilar regions, anatomically consistent with the expected distribution of pulmonary TB. This provides evidence that the model has learned *clinically meaningful* features, not purely image statistics.

For false negative cases, the attention maps are diffuse or focus on incorrect regions (diaphragm, cardiac silhouette), suggesting the model failed to register the pulmonary pathology. Possible causes include subtle or early-stage infiltrates that are also difficult for radiologists to detect (explaining potential label noise in this subset).

For false positive cases, attention frequently lands on rib shadows, reticular opacities from aging or prior infection, or imaging artefacts such as clothing buttons, illustrating that the model has not perfectly separated TB-specific from incidental CXR findings.

Chapter 3
Discussion

3.1. Revisiting the Research Questions

Primary RQ: Can FL+DP match centralised performance? The answer is: **not in this experiment, but not definitively no either.** The FL+DP model (AUC = 0.730) fell significantly short of the centralised baseline (AUC = 0.894; McNemar $p = 0.012$) and did not meet the pre-registered success criteria. However, the shortfall is primarily attributable to a technical failure rather than a fundamental ceiling of FL+DP on this task. The FL (No DP) model (AUC = 0.909) exceeded the centralised baseline, establishing that the federated learning framework *without* DP is fully viable.

SQ1: How does performance degrade with decreasing ε? The DP sweep produced incomplete results due to the inplace autograd bug at some epsilon settings. The available evidence suggests that $\varepsilon = 8$ already produces substantial noise at small site sizes (UCH Ibadan: $\sigma = 1.75$). A clean re-run is needed to characterise the full ε–AUC trade-off curve.

SQ2: How does α affect FL convergence? The ablation shows that performance degrades dramatically at $\alpha = 0.1$ (AUC = 0.385) due to extreme client drift, and non-monotonically at $\alpha = 1.0$ (AUC = 0.659) due to a specific adverse Dirichlet draw. At $\alpha = 0.5$, performance reaches AUC = 0.818. The non-monotonic $\alpha = 1.0$ result is a cautionary tale: the Dirichlet concentration parameter does not uniquely determine performance; the specific partition matters, and reporting α alone is insufficient.

SQ3: How many rounds for convergence? The round ablation shows convergence by Round 2 (AUC = 0.876, stable at Round 3: 0.876). The rapid convergence is driven by the ImageNet initialisation; for training-from-scratch FL, many more rounds would be required.

SQ4: Are errors equitably distributed across sites? No. The equity gap of 0.113 AUC units (Lagos 0.952, OAU 0.839) indicates that patients at different simulated sites experience substantially different model performance. OAU's low sensitivity (0.667) is particularly concerning.

3.2. Interpreting the FL (No DP) Outperformance

The finding that FL (No DP) achieves higher AUC (0.909) than the centralised model (0.894) on the same test set merits careful interpretation. The confidence intervals overlap substantially ([0.853, 0.953] vs [0.833, 0.943]), so this difference is within sampling noise. Nevertheless, there are principled reasons why FL can sometimes match or exceed centralised performance:

1. **Implicit regularisation through heterogeneity.** Each site's local training pushes the global model in a different direction, and FedAvg averages over these diverse gradients. This can function as a regulariser, preventing overfitting to the idiosyncrasies of any single site.
2. **Ensemble effect.** FedAvg weight averaging produces a model that interpolates between the local optima of all clients. For convex loss surfaces, this is optimal; for non-convex neural networks, it can still land in a flat, well-generalising basin.[1]
3. **Operating point shift.** FL (No DP) has higher specificity and lower sensitivity than centralised; the AUC advantage reflects the entire ROC curve, not just the operating point. At the clinical operating point (maximising sensitivity), centralised is preferred.

3.3. Comparison with Prior Work

Our FL (No DP) AUC of 0.909 is comparable to Adnan, Kalra, Cresswell, Taylor, and Tizhoosh (2022)'s best FL (No DP) result of 0.91, and our FL+DP result of 0.730 is below their DP result of 0.83 at $\varepsilon = 8$.

The difference in FL+DP performance is explained by:

1. Smaller dataset ($n = 800$ vs their larger CheXpert subset)
2. Client failures due to the inplace autograd bug
3. Smaller sites (UCH Ibadan: $n = 53$) requiring higher noise multipliers ($\sigma = 1.75$)

Our stricter evaluation (bootstrap CIs, McNemar's test, non-inferiority analysis) distinguishes this work from prior results that report only point estimates.

[1] The geometric interpretation of FedAvg as interpolation between local optima and its relationship to loss surface geometry is discussed in Li, Sahu, Talwalkar, and Smith (2020).

Chapter 4

Limitations

4.1. Dataset Limitations

4.1.1. Non-Nigerian Source Data

The most consequential limitation of this project is that *no Nigerian-origin data were used*. The Montgomery County dataset was collected in Maryland, USA (2012–2014) on U.S. patients with U.S.-standard X-ray equipment. The Shenzhen Hospital dataset was collected in Guangdong, China on Chinese patients.

4.1.2. Small Dataset

$n = 800$ total images is very small by modern deep learning standards. **Consequences:**

- Wide bootstrap confidence intervals that prevent strong statistical conclusions.
- Risk of overfitting, mitigated but not eliminated by transfer learning and dropout.
- Per-site test sets as small as 17 images (Aminu Kano), making per-site metrics highly variable.

4.1.3. Label Noise

Radiologist-assigned labels carry unquantified noise. There is no inter-rater reliability statistic for either dataset. The 11 false negatives with mean confidence 0.098 might reflect genuinely subtle TB presentations *or* mislabelled normal images *or* a combination. Without laboratory confirmation (sputum culture) for all 800 images, we cannot distinguish these cases.

4.2. Federated Learning Simulation Limitations

4.2.1. Simulated vs Real Hospital Sites

The five sites are mathematical constructs from a Dirichlet partition of two source datasets. They share a common image distribution (Montgomery + Shenzhen), whereas real hospitals would produce images with different equipment signatures. The within-site data are not actually *local* to any physical hospital; the partition is only a statistical approximation of between-hospital heterogeneity.

Real hospitals would additionally exhibit:

- Temporal heterogeneity (seasonal TB incidence variation)
- Demographic heterogeneity (age distribution, sex ratio, HIV prevalence differ by catchment population)
- Labelling heterogeneity (different radiologists apply different diagnostic thresholds)
- Equipment heterogeneity (CXR machine brand, detector type, exposure protocol)

None of these are captured by Dirichlet partitioning.

4.3. Differential Privacy Limitations

4.3.1. Inplace Autograd Incompatibility

The Opacus inplace ReLU bug caused partial client failures in the FL+DP mock run, making the FL+DP results unreliable. The fix (setting `inplace=False` in all ReLU activations) is straightforward but was not applied in time for the reported results. The FL+DP AUC of 0.730 should be treated as a lower bound, not a true characterisation of the system's capability.

4.3.2. Sequential Composition Is Conservative

Our per-round budget allocation ($\varepsilon_r = \varepsilon/R$) uses the basic sequential composition theorem, which is loose. Tighter analysis using the moments accountant or Rényi DP composition across rounds could achieve the same total ε with less noise per round, potentially improving utility.

4.3.3. $\varepsilon = 8$ Is a Weak Privacy Guarantee

In the formal DP literature, $\varepsilon = 8$ is considered a pragmatic but not conservative choice. Gradient inversion attacks have been demonstrated to partially recover training images from gradient updates even with moderate DP noise. For truly sensitive medical data, $\varepsilon \leq 2$ or $\varepsilon \leq 1$ is often recommended, with corresponding steeper utility costs.[1]

4.3.4. BatchNorm–GroupNorm Confound

The DP model uses GroupNorm while the non-DP models use BatchNorm. This confound means the performance gap between DP and non-DP models reflects both noise addition *and* normalisation layer differences. A fair ablation would compare FL+DP (GroupNorm) vs FL+DP (BatchNorm) vs FL (No DP, GroupNorm) to isolate each effect.

[1] The state-of-the-art in gradient inversion attacks and their interaction with DP is surveyed in Geiping et al. (2020), "Inverting Gradients – How easy is it to break privacy in federated learning?" at https://arxiv.org/abs/2003.14053.

4.4. Evaluation Limitations

- **No external validation set.** All results are from a held-out subset of the same Montgomery/Shenzhen distribution. External validation on an independent Nigerian dataset is essential before drawing conclusions about real-world performance.
- **Wide confidence intervals.** With $n = 120$ test images, the 95% bootstrap CI on AUC has width ≈ 0.11. Results that appear different may be compatible within sampling error.
- **Symmetric Youden threshold.** The clinical cost function for TB screening is asymmetric (FN cost $\gg$ FP cost). A risk-adjusted threshold would increase sensitivity at the cost of specificity.
- **No calibration analysis.** The model's predicted probabilities may not be well-calibrated (i.e., a predicted probability of 0.8 may not correspond to 80% empirical TB prevalence among such cases). Reliability diagrams and calibration error analysis are needed before clinical use.

Chapter 5

Conclusions and Future Work

5.1. Summary of Findings

Finding 1: FL without DP matches centralised performance. The federated model trained without differential privacy achieved AUC-ROC = 0.909 [95% CI: 0.853–0.953], numerically exceeding the centralised baseline (0.894 [0.833–0.943]), while sharing zero raw patient images across sites. This finding demonstrates technical feasibility: a FedAvg network across five heterogeneous simulated hospitals delivers performance indistinguishable from centralised training.

Finding 2: FL+DP did not meet the success criteria in this experiment. The FL+DP model achieved AUC-ROC = 0.730 [95% CI: 0.642–0.815], significantly below the centralised baseline (McNemar $p = 0.012$; non-inferiority not established). This failure is primarily technical (Opacus inplace autograd incompatibility causing partial client failures), not a fundamental ceiling; a clean re-run is expected to improve results substantially.

Finding 3: FL converges within 2 communication rounds. The round ablation showed convergence by Round 2 (AUC = 0.876), driven by the strong ImageNet initialisation. In a real Nigerian hospital deployment with 10 Mbps connectivity, 2 rounds of FedAvg would take $\approx$ 3 minutes total—a negligible operational overhead.

Finding 4: The equity gap of 0.113 requires attention. Per-site analysis revealed an AUC equity gap of 0.113 units between the best (Lagos, AUC = 0.952) and worst (OAU, AUC = 0.839) simulated sites. OAU's sensitivity of 0.667 is clinically unacceptable. Per-site threshold calibration is mandatory before deployment.

Finding 5: The model learns genuine TB features. The shuffled-label control (AUC = 0.447 $\approx$ 0.5) and GradCAM anatomical attention maps together provide moderate evidence that the model has learned clinically meaningful features rather than exploiting spurious shortcuts.

5.2. Recommendations for Future Work

Immediate (prerequisite before any clinical use):

1. **Fix the Opacus inplace ReLU bug.** Apply `inplace=False` to all ReLU activations in the ResNet-18 definition and re-run the full FL+DP 50-round experiment.
2. **Upgrade to Flower 2.x.** The current codebase uses a deprecated Flower API. The Flower 2.x simulation API provides improved performance and more reliable client management.
3. **Implement per-site threshold calibration.** Use Platt scaling on each site's local validation set to produce calibrated probabilities and a site-specific threshold that maximises sensitivity at an acceptable false positive rate.

Near-term (within 6 months):

4. **Collect Nigerian-origin CXR data.** Partner with Lagos University Teaching Hospital, UNTH Enugu, and Aminu Kano Teaching Hospital to collect and annotate ≥ 500 CXRs per site under IRB approval. This is the single highest-impact action for improving clinical validity.
5. **Implement leave-one-site-out evaluation.** Train on $K-1$ sites and test on the held-out site to directly assess cross-site generalisation—the most clinically relevant metric.
6. **Apply tighter DP accounting.** Use Rényi DP composition across FL rounds (instead of sequential composition) to achieve the same total ε with less per-step noise.
7. **Conduct the GroupNorm ablation.** Train FL (No DP) with GroupNorm to isolate the normalisation confound from the DP noise effect.

Longer-term (research directions):

8. **Explore FedProx and FedNova.** These FL algorithms add proximal regularisation to local updates (FedProx) or normalise local update magnitudes (FedNova) to reduce client drift in heterogeneous settings. Both may outperform FedAvg at our level of data heterogeneity.[1]
9. **Add metadata and demographic fairness analysis.** Collect age, sex, and HIV status for each patient to enable subgroup fairness analysis per the framework of Barocas, Hardt, and Narayanan (2023).
10. **Investigate larger architectures.** DenseNet-121 or EfficientNet-B4 operating at 320×320 may improve sensitivity on subtle TB presentations, at the cost of larger DP noise requirements.

[1] FedProx is described in Li et al. (2020b), available at https://arxiv.org/abs/1812.06127. FedNova is described in Wang et al. (2020) at https://arxiv.org/abs/2007.07481.

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